Chapter 4: Interest Rates (Part 1)

> **Context & Scope:**
>
> - Interest rates are central to valuing virtually all derivatives.
> - **Day Count Conventions:** For ease of exposition, this chapter ignores day count conventions (to be covered in Chapters 6 & 7).
> - **Continuous Compounding:** Used extensively in this book for derivative analysis.

4.1 Types of Rates

An interest rate defines the promised payment amount from a borrower to a lender. The specific rate depends heavily on **Credit Risk** (the risk of default).

- **General Rule:** Higher Credit Risk $\rightarrow$ Higher Promised Interest Rate.
- **Unit of Measurement: Basis Points**.
  - 1 Basis point = 0.01% per annum.

1. Treasury Rates

Rates an investor earns on government bills and bonds.

- **Definition:** The rate at which a government borrows in its own currency.
  - *Example:* Japanese Treasury rates for Yen borrowing; US Treasury rates for Dollar borrowing.
- **Risk Profile:** Theoretically **Total Risk-Free**.
  - *Assumption:* A government will not default on obligations in its own currency (it can print money).

2. LIBOR (London Interbank Offered Rate)

An **unsecured** short-term borrowing rate between banks.

- **Usage:** Reference rate for hundreds of trillions of dollars in transactions (e.g., Interest Rate Swaps).
- **Tenor:** Quoted for 15 borrowing periods (1 day to 1 year) across 10 currencies.
- **Calculation Method:**
  - Published by the **British Bankers Association (BBA)** at 11:30 AM (UK time).
  - Banks provide estimates of their borrowing costs just prior to 11:00 AM.
  - **Trimming:** The top 25% and bottom 25% of quotes are discarded.
  - **Average:** The remaining quotes are averaged to fix the rate.
- **Credit Rating:** Corresponds to **AA-rated** financial institutions (AA is the second-best rating after AAA).
- **Manipulation & Issues:**
  - *Reason 1:* Banks manipulated quotes to appear healthier (lower borrowing costs = financial strength).
  - *Reason 2:* To profit from derivatives tied to LIBOR fixings.
  - *Future:* Likely to be replaced by rates based on actual transactions due to lack of liquidity in interbank borrowing.

3. The Fed Funds Rate

The overnight interest rate for interbank lending in the United States.

- **Mechanism:**
  - US banks must maintain specific cash **reserves** with the Federal Reserve based on their assets/liabilities.
  - Banks with **surplus** reserves lend to banks with **deficits** overnight.
- **Effective Federal Funds Rate:** The **weighted average** of rates in brokered transactions (weighted by transaction size).

- **Central Bank Role:** The Fed monitors this rate and intervenes to raise or lower it.
- **International Equivalents:**
  - **SONIA:** Sterling Overnight Index Average (UK).
  - **EONIA:** Euro Overnight Index Average (Euro Zone).
- **LIBOR vs. Fed Funds:**
  - Both are **unsecured**.
  - **Spread:** Overnight LIBOR is typically **6 basis points (0.06%) higher** than the Fed Funds rate.
  - *Reasons for spread:* Timing effects, borrower composition differences (London vs. New York), and settlement mechanisms.

4. Repo Rates (Repurchase Agreements)

A **secured** borrowing rate.

- **Mechanism:**
  i. A financial institution owning securities agrees to **sell** them for price X.
  ii. It agrees to **buy them back** later for a slightly higher price Y.
  iii. **Interest:** The difference $(Y - X)$ represents the interest paid on the loan.
- **Risk Profile:** Very little credit risk because it is collateralized.
  - If borrower defaults $\rightarrow$ Lender keeps the securities.
  - If lender defaults $\rightarrow$ Original owner keeps the cash.
- **Comparison:** Repo rates are generally **slightly lower** than the Fed Funds rate (because they are secured).
- **Types:**
  - *Overnight Repo:* Rolled over day-to-day (most common).
  - *Term Repo:* Longer-term arrangements.

5. The "Risk-Free" Rate

A theoretical concept essential for valuing derivatives (risk-neutral valuation).

- **The Proxy Problem:** There is no perfect "risk-free" rate.
- **Historical Proxy:** LIBOR was traditionally used, even though AA banks have a small chance of default.
- **Current State:** Practitioners are shifting away from LIBOR to other proxies (discussed in Chapter 9) as the concept of "risk-free" is refined.

4.2 Measuring Interest Rates

The meaning of an interest rate quote is ambiguous without a **Compounding Frequency**.

- **Analogy:** The difference between compounding frequencies is like the difference between **kilometers and miles**. They are different units for measuring the same phenomenon.

Effect of Compounding Frequency

Scenario: $100 invested for 1 year at a quoted rate of 10%.

Compounding Frequency (m)	Calculation	Value at End of Year
Annual $(m = 1)$	100×1.10	**$110.00**
Semiannual $(m = 2)$	$100 \times 1.05 \times 1.05$	**$110.25**
Quarterly $(m = 4)$	100×1.025^4	**$110.38**

Compounding Frequency (m)	Calculation	Value at End of Year
Monthly $(m = 12)$	100×1.00833^{12}	**$110.47**
Weekly $(m = 52)$	100×1.00192^{52}	**$110.51**
Daily $(m = 365)$	$100 \times (1 + \frac{0.10}{365})^{365}$	**$110.52**
Continuous $(m \to \infty)$	$100 \times e^{0.10}$	**$110.52**

Formulas for Valuation

If amount **A** is invested for **n** years at rate **R**:

1. General Compounding (m times per annum):

$$\text{Terminal Value} = A \left(1 + \frac{R}{m}\right)^{mn}$$

2. Continuous Compounding:

$$\text{Terminal Value} = Ae^{Rn}$$

(Note: $e \approx 2.71828$. This is built into most calculators).

> **Practical Note:** For most purposes, continuous compounding is equivalent to daily compounding. It is the standard for derivatives pricing in this book.

Conversion Formulas

We can convert between a rate compounded m times per annum (R_m) and a continuously compounded rate (R_c).

To convert TO Continuous (R_c):

$$R_c = m \ln \left(1 + \frac{R_m}{m}\right)$$

To convert FROM Continuous (R_m):

$$R_m = m \left(e^{R_c/m} - 1\right)$$

Examples from Text

Example 4.1: Converting to Continuous

- **Given:** 10% per annum, Semiannual compounding ($m = 2, R_m = 0.10$).
- **Calculation:**

$$2 \ln \left(1 + \frac{0.10}{2}\right) = 2 \ln(1.05) = 0.09758$$

- **Result:** 9.758% per annum (continuously compounded).

Example 4.2: Converting from Continuous

- **Given:** 8% per annum Continuous ($R_c = 0.08$), Interest paid Quarterly ($m = 4$).
- **Calculation:**

$$4\left(e^{0.08/4} - 1\right) = 4(e^{0.02} - 1) = 0.0808$$

- **Result:** 8.08% per annum (quarterly compounding).
- **Implication:** On a $1,000 loan, required quarterly payments are $20.20($1000 \times 0.0808 / 4$).

4.3 Zero Rates

Also known as **Spot Rate**, **Zero Rate**, or **Zero**.

- **Definition:** The n-year zero-coupon interest rate is the rate earned on an investment that starts today and lasts for n years.
- **Key Feature:** All interest and principal are realized at the **end** of n years. There are **no intermediate payments**.

Calculation Example:

- **Given:** 5-year zero rate is 5% per annum (continuous).
- **Investment:** $100.
- **Future Value:**

$$100 \times e^{0.05 \times 5} = 100 \times e^{0.25} = 128.40$$

Relationship to Bonds:
Most market rates (like a 5-year Treasury bond with a 6% coupon) are **not** zero rates because the investor receives cash (coupons) *before* the end of year 5.

- *Note:* The text implies a method (Bootstrap Method) will be discussed later to extract Zero Rates from these coupon-bearing bonds.